Roll No. .....

TEE–401

B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

ELECTRICAL MACHINES–II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the capacitor start and capacitor run motor with suitable diagram. (CO1, CO3)

(b) Draw the equivalent circuit of single phase induction motor and obtain the equation for mechanical power developed. (CO1, CO3)

(c) A 2.5-kW 120-V 60-Hz capacitor-start motor has the following impedances for the main and auxiliary windings (at starting) : (CO1, CO3)

$Z_{main} = 4.5 + j\,3.7\ \Omega$ main winding

$Z_{aux} = 9.5 + j\,3.5\ \Omega$ auxiliary winding

Find the value of starting capacitance that will place the main and auxiliary winding currents in quadrature at starting.

P. T. O.

2. (a) Discuss torque slip characteristics of three phase Induction motor. (CO2, CO6)

(b) A 460-V, 60-Hz, 80-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2 kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities : (CO2, CO6)

(i) The air-gap power P_{AG}

(ii) The power converted P_{Conv}

(iii) The output power P_{olitt}

(iv) The efficiency of the motor

(c) Draw equivalent circuit of three induction motor and hence find the expression for gross mechanical power developed. (CO2, CO6)

3. (a) With the help of suitable diagram discuss the role of energy and co-energy in detail. (CO4, CO6)

(b) Discuss three phase rotating magnetic field with suitable diagram. (CO4, CO6)

(c) Obtain the equation of torque for multiply excited magnetic field system using co-energy. (CO4, CO6)

4. (a) The full load current of a 3.3 kV star connected synchronous motor is 160 A at 0.8 pf lagging. The resistance and reactance of the motor are 0.8 Ω and 5.5 Ω per phase respectively. Calculate the excitation emf, torque angle, efficiency and shaft output of the motor assume mechanical stray losses to be 30 kW. (CO4, CO5)

(b) Discuss Open Circuit Characteristics and Shot Circuit Characteristics of synchronous generator. (CO4, CO5)

(c) Derive the expression of power angle characteristics and discuss it in details. (CO4, CO5)

5. (a) A 3300V, delta connected motor has synchronous resistance per phase (delta) of 10 Ω. It operates at leading power factor of 0.707 when drawing 800 kW from mains. Calculate excitation emf. (CO6, CO5)

(b) The nameplate on a 460-V, 50-hp, 60-Hz, four-pole induction motor indicates that its speed at rated load is 1755 r/min. Assume the motor to be operating at rated load. (CO6, CO5)

(i) What is the slip of the rotor ?

(ii) What is the frequency of the rotor currents ?

(iii) What is the angular velocity of the stator-produced air-gap flux wave with respect to the star ? With respect to the rotor ?

(c) Explain the double-revolving-field theory of single-phase Induction Motors. (CO6, CO5)

Roll No. .....

TMA–401

B. TECH. (EEE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

MATHEMATICS—III

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) If 12 persons are seated at a round table, what is the probability that two particular persons sit together ? If 6 boys and 6 girls sit in a row randomly, find the probability that all the 6 girls sit together. (CO1)

(b) It is known that 4% of a batch of components in a manufacturing process are faulty. Component are tested on the production line with 90% probability that a fault component is detected, but it is known that in 2% of the cases a component which is not faulty is nevertheless recorded as faulty. What is the probability that a component which is recorded as faulty is actually faulty ? (CO1)

P. T. O.

(c) At an electrical company, it is known from past experience that the probability a worker who attended the company's program meets the production quato is 0.90. The corresponding probability for new worker who did not attend the training program is 0.25. It is also known that 80% of all new worker attend the company's program. Find the probability that a new worker who met the production quota would have attended the company's training program. (CO1)

2. (a) Let X be a continuous random variable with probability density function :

$$f(x)=\begin{cases} kx & ;\ 0\le x\le 1 \\ k & ;\ 1\le x\le 2 \\ -kx+3k & ;\ 2\le x\le 3 \\ 0 & ;\ \text{elsewhere} \end{cases}$$

(i) Determine the value of K.

(ii) Mean and variance of X. (CO2)

(b) Suppose the temperature T during May is normally distributed with Mean 68 and Standard Deviation 6. Find the probability P that the temperature during May is :

(i) between 70° and 80°.

(ii) less than 60°. (CO2)

(c) Let X be random variable having Binomial distribution with parameter. $n=25$ and $p=\frac{1}{5}$. Evaluate $P[X<\mu_x-2\sigma_x]$, where μ_x is Mean σ_x is Standard deviation . (CO2)

3. (a) The joint density of X and Y is $f(x,y)=\frac{x+2y}{27}$, where x and y can be assumed only integer value 0, 1 and 2. Find the conditional distribution of Y for $X=x$. (CO3)

(b) If X and Y have join p.d.f.

Y \ X	1	2	3
1	0.1	0.1	0.2
2	0.2	0.3	0.1

Find (i) Marginal distribution of X and Y, (ii) Conditional distribution of X given Y =1 and (iii) Evaluate $P[x+y<4]$. (CO3)

(c) Toss 3 coins. Let X denotes the number of head on the first two and Y denotes the number of heads on the last two.

(i) Find the joint distribution of X and Y.

(ii) Find $E[y/x=1]$.

(iii) Find the correlation coefficient $\rho(x,y)$. (CO3)

4. (a) To study the relationship between Aptitude test scores and productivity resulting after three months training of workers in a factory, a survey was conducted by a researcher, who had taken a sample of 9 workers :

Aptitude Scope	Productivity
1	9
3	10
5	11
7	14
9	15
2	8
4	12
6	13
8	16

Calculate coefficient of correlation between Aptitude test scores and Productivity and give your comments. (CO4)

(b) A garage has recorded the following faults in cars of four makes over a period of time :

Make	Nature of faults		
	Mech.	Electrical	Chasis
A	32	26	12
B	60	54	22
C	52	38	15
D	28	25	12

Can you say the type of fault is independent of the make of the car ?

(Given $\chi^2 = 12.592$, for 6 degrees of freedom). (CO4)

(c) Two samples are drawn from two normal populations. From the data given below, test whether the two samples have the same variance at 5 % level of significance :

Sample I	Sample II
60	61
65	66
71	67
74	85
76	78
82	63
85	85
87	86
-	88
-	91

Tabulated value at 5% level of significance and 9 degree of freedom is 3.68.

(CO4)

5. (a) In Miss India Contest ten competitors are ranked by three judges in the given table. Use the correlation coefficient to determine which pair judge has the nearest approach to common perception in beauty : (CO5)

1st Judge	2nd Judge	3rd Judge
5	1	6
3	6	4
10	5	9
7	10	8
2	3	1
1	2	2
4	4	3
10	9	10
4	7	5
6	8	7

(b) A random sample of 16 values from a normal population showed a mean of 41.5 inches and a sum of square of deviation from this mean equal to 135 inches. Show that the assumption of a population mean of 43.5 inches is not valid at 5% level of significance. (For two-tailed test $t_{15(0.025)} = 2.13$). (CO5)

(c) Calculate linear regression coefficients from the following : (CO5)

x	y
1	3
2	7
3	10
4	12
5	14
6	17
7	20
8	24

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Roll No. .....

TEE–402

B. TECH. (EE) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

INTRODUCTION TO ELECTRICAL ENERGY SOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write a short note on 'Electricity Cost Reduction'. (CO1)

   (b) Why do modern steam turbines have a number of stages ? (CO1)

   (c) Discuss the characteristics of a base load power plant. Give suitable examples. Differentiate between fixed and operating costs of a power plant. (CO1)

2. (a) What considerations must be kept in view in siting nuclear plants ? (CO2)

   (b) Discuss the fuel cycle for a nuclear plant. (CO2)

   (c) Find the energy equivalent of 1 gram of matter. Define mass defect and binding energy. (CO2)

3. (a) How are hydroturbines classified ? (CO3)

P. T. O.

(b) A hydroelectric station is designed to operate at a mean head of 205 m and fed by a reservoir having a catchment area of 1000 km. with an annual rainfall of 125 cm of which 80% is available for power generation. The expected load factor is 75%. Allowing a head loss of 5 m and assuming efficiency of turbine and generator to be 0.9 and 0.95, calculate suitable MW rating of the station. Comment on the type of turbine to be used. (CO3)

(c) How are hydroplants classified ? (CO3)

4. (a) Discuss the advantages, disadvantages and fields of application of Diesel electric plants. (CO4)

(b) Discuss the main parts of a diesel power plant. (CO4)

(c) Discuss the working of Gas Turbine Plant with the help of a suitable diagram. (CO4)

5. (a) The load curve of an electrical system is linear with the following values at different times of the day : (CO5, CO6)

Time	Load (MW)
12	50
5 a.m.	50
9	100
6	100
8 p.m.	150
10	80
12	50

Plot chronological load curve. Find the energy required by the system in one day and the system daily load factor.

(b) Define Tariff. Why do some power companies put a penalty for lower power factors ? (CO5, CO6)

(c) Incremental fuel cost in rupees per Megawatt hour for two units in a plant are given by following two equations :

$$0.1P_1 + 20 \text{ and } 0.12P_2 + 16$$

The minimum and maximum load on each unit is to be 20 MW and 125 MW. Determine incremental fuel cost and allocation of load between units for minimum cost when load is 100 MW. (CO5, CO6)

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Roll No. .....

TEE–403

B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

SIGNALS & SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain periodic and aperiodic signals. Also, explain the condition for the sum of two periodic signals to be periodic. (CO1)

(b) Determine energy and power of the following signals : (CO1)

(i) $x(t) = tu(t)$

(ii) $x(t) = e^{-3t}u(t)$.

(c) Prove the following properties of continuous-time unit impulse function : (CO1)

(i) Shifting Property

(ii) Scaling Property

(iii) Differentiation Property.

P. T. O.

2. (a) Discuss the following systems : (CO2)

   (i) Linear and non-linear systems

   (ii) Time invariant and time variant systems.

   (b) If convolution of two signals, $x_1(t)$ and $x_2(t)$ is $z(t)$, i.e., $z(t) = x_1(t) * x_2(t)$, then show that : (CO2)

   (i) $z(t-\mathrm{T}) = x_1(t) * x_2(t-\mathrm{T}) = x_1(t-\mathrm{T}) * x_2(t)$

   (ii) $z(t) = x_1(t) * x_2(t) = x_2(t) * x_1(t)$.

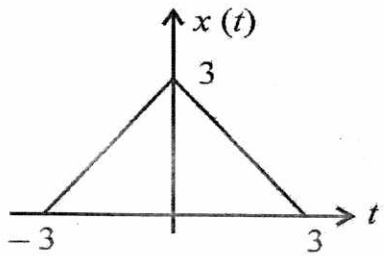
   (c) Determine the convolution continuous-time signals : (CO2)

   (i) $x_1(t) = e^{-at}u(t)$; and $x_2(t) = e^{-bt}u(t)$

   (ii) $x_1(t) = u(t)$; and $x_2(t) = u(t)$.

3. (a) Derive the expression of Fourier transform of signal $x(t)$ using the exponential form of the Fourier series. (CO3)

   (b) Determine the Fourier transform of the triangular signal shown below : (CO3)



   (c) If Fourier transform of the signal $x(t)$ is $\mathrm{X}(\omega)$, then show that : (CO3)

   (i) The Fourier transform of $x(at)$ will be $\frac{1}{(a)}\mathrm{X}\left(\frac{\omega}{a}\right)$

   (ii) The Fourier transform of $\frac{dx(t)}{dt}$ will be $j\omega\mathrm{X}(\omega)$.

4. (a) Find the Laplace transform of the following signals : (CO4)

   (i) $x(t) = e^{-2t}u(t-2)$

   (ii) $x(t) = e^{-t}\cos(\omega t + \varphi)u(t)$.

(b) If Laplace transform of the signal $x(t)$ is X(s) then find the Laplace transform of the following signals : (CO4)

(i) $tx(t)$

(ii) $x(at)$.

(c) Find the inverse Laplace transform of the following : (CO4)

(i) $X(s) = \dfrac{s}{s^2 + 5s + 6}$

(ii) $X(s) = \dfrac{3s^2 + 8s + 6}{(s^2 + 2s + 1)(s + 2)}$.

5. (a) Discuss sampling theorem. (CO5, CO6)

(b) Determine the Nyquist rate of the following signals : (CO5, CO6)

(i) $x(t) = \sin 400\pi t$

(ii) $x(t) = \sin^3 \pi t$

(iii) $x(t) = \cos(150\pi t) \times \sin(100\pi t)$.

(c) If Laplace transform of $x_1(t)$ and $x_2(t)$ are $X_1(s)$ and $X_2(s)$, then show that Laplace transform of their convolution will be the product of $X_1(s)$ and $X_2(s)$. (CO5, CO6)

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TEE–404

B. TECH. (EE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

DIGITAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Design a 4-bit binary (B_3, B_2, B_1, B_0) to Gray (G_3, G_2, G_1, G_0) code converters circuit. (CO1)

(b) Write the basic Boolean laws and their respective Boolean expressions. (CO1)

(c) Find the minimum numbers of two input NAND gates required to implement Boolean function Y = A B–C, assuming that A, B and C are available. (CO1)

2. (a) Simplify the Boolean function "*f*" using the don't-care conditions *d* in : (CO2, CO3)

(i) Sum of Product and

(ii) Product of Sum forms :

$$f(w,x,y,z) = \Sigma m\ (0, 1, 6, 7, 14, 15) + \Sigma d\ (3, 4, 11, 12)$$

P. T. O.

(b) Write the De Morgan's theorems and prove both of them. (CO2, CO3)

(c) Find the number of min-terms in the following function : (CO2, CO3)

$$F(w, x, y, z) = y' + w'z' + xz'$$

3. (a) Design a logic circuit that has three inputs A, B and C and whose output "Y" is high (1) only when a majority of the input is high. (CO3, CO4)

(b) What is magnitude comparator ? Draw the logic circuit diagram of a single-bit comparator. Describe its working. (CO3, CO4)

(c) Define full adder, show its truth table and derive logical expressions for sum and carry. (CO3, CO4)

4. (a) Discuss triggering of flip-flops. Draw the symbols of the following flip-flops : (CO5)

(i) T-flip-flop and its truth table

(ii) Low-level triggered JK-flip-flop

(iii) Leading-edge triggered JK-flip-flop

(iv) Training-edge triggered JK-flip-flop.

(b) Draw the logical circuit of master-slave flip-flop. Describe its working and usefulness. (CO5)

(c) Convert a D-flip-flop into a T-flip-flop. Show combined conversion table that includes characteristics table and excitation table. Drive the required conversion logic. (CO5)

5. (a) Design a Mod 3 synchronous converter. (CO6)

(b) Draw the circuit diagram of a Mod 10 Asynchronous counter using T-flip-flops. Draw waveshapes of all the Q-line outputs of all the flip-flops. (CO6)

(c) What do you understand by the "present and clear state" inputs of flip-flop ? Discuss their usefulness in Mod M counters. (CO6)